Expansion Methods in Inequalities

Version 1.0

16-7-2022

Contents

Introduction

This note focuses on some expansion methods in solving inequality problems, including completing squares, Muirhead's theorem, Schur's inequality, the *uvw* method, etc. The advantage of such methods is that, during expansion, we do not change the bound of the desired inequality. So the inequality must be true under the assumption that the problem is true and the expansion is correct! Very often, we do not need to waste time in thinking of a correct way to solve the problem. The only issue is that the intermediate steps are tedious.

However, MO problems in recent years are set in a way which is against such expansion methods. Therefore, this note is more for fun than for practical use.

1 Completing Squares and Trivial Bounds

The most basic inequality satisfied by every real number x is probably the inequality $x^2 \geqslant 0$. Every inequality about real numbers essentially comes from this fact, and possibly together with other trivial bounds which are given (e.g. if x is a positive real number, then $x > 0$). In view of this, one strategy in proving inequalities is to expand everything, then group the terms in some ways and complete the proof by using these trivial facts.

We first begin with a simple but important result.

Problem 1.1. Let a, b, c be real numbers. Prove that

$$a^2 + b^2 + c^2 \geqslant ab + bc + ca.$$

Solution. This follows from

$$a^2 + b^2 + c^2 - (ab + bc + ca) = \frac{1}{2}\left((a-b)^2 + (b-c)^2 + (c-a)^2\right) \geqslant 0.$$

$\square$

If this sounds too simple, let us try the next one.

Problem 1.2. Let a, b, c be real numbers. Prove that

$$(a^2 + b^2 + c^2)^2 \geqslant 3(a^3b + b^3c + c^3a).$$

Solution. This follows from

$$(a^2 + b^2 + c^2)^2 - 3(a^3b + b^3c + c^3a) = \frac{1}{2}\sum_{\text{cyc}}(a^2 - 2ab + ac + bc - c^2)^2 \geqslant 0.$$

$\square$

The above example is much harder than one may imagine, because there are equality cases other than $a = b = c$, such as $(a, b, c) = \left(\sin^2\frac{4\pi}{7}, \sin^2\frac{2\pi}{7}, \sin^2\frac{\pi}{7}\right)$.

Completing squares can be used even if there are n variables.

Problem 1.3. (Cauchy-Schwarz inequality) Let $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_n$ be real numbers. Prove that

$$(a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2) \geqslant (a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2.$$

Solution. The result is obvious if $a_1 = a_2 = \cdots = a_n = 0$. Otherwise, let

$$x = \frac{a_1b_1 + a_2b_2 + \cdots + a_nb_n}{a_1^2 + a_2^2 + \cdots + a_n^2}.$$

It is routine to check that

$$\left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right) - \left(\sum_{i=1}^n a_ib_i\right)^2 = \left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n (a_ix - b_i)^2\right) \geqslant 0.$$

The result follows readily. $\square$

The above Cauchy-Schwarz inequality can be proved using completing square. How about another well-known result — the AM-GM inequality? First of all, we cannot merely use completing square because we have an extra condition that the variables are positive. Thus, we probably need to use this trivial bound in the proof. Secondly, it seems like there is not any universal way to rewrite the expression as a sum of nonnegative terms that works for every n. However, for any given n, we are able to do so. In particular, when n is a power of 2, it is easy to see from the well-known proof that it is possible to prove the inequality by completing squares. For other values of n, we can also do similar things.

Problem 1.4. Let a, b, c be positive real numbers. Prove that

$$a^3 + b^3 + c^3 \geqslant 3abc.$$

Solution. This is because

$$a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a + b + c)\big((a - b)^2 + (b - c)^2 + (c - a)^2\big) \geqslant 0.$$

$\square$

Problem 1.5. Let a, b, c, d, e be positive real numbers. Prove that

$$\frac{a + b + c + d + e}{5} \geqslant \sqrt[5]{abcde}.$$

Solution. Let $x = \dfrac{a + b + c + d + e}{5}$ and $y = \sqrt[5]{abcde}$. Firstly, we note that

$$\begin{aligned} x^{\frac{5}{6}} - y^{\frac{5}{6}} &= \frac{\begin{aligned}&(\sqrt{a} - \sqrt{b})^2 + (\sqrt{c} - \sqrt{d})^2 + (\sqrt{e} - \sqrt{x})^2 \\ &+ (\sqrt[6]{ab} + \sqrt[6]{cd} + \sqrt[6]{ex})\big((\sqrt[6]{ab} - \sqrt[6]{cd})^2 + (\sqrt[6]{cd} - \sqrt[6]{ex})^2 + (\sqrt[6]{ex} - \sqrt[6]{ab})^2\big)\end{aligned}}{6x^{\frac{1}{6}}} \\ &\geqslant 0. \end{aligned}$$

Then we easily obtain

$$x - y = \frac{(x^{\frac{5}{6}} - y^{\frac{5}{6}})(x^{\frac{25}{6}} + x^{\frac{10}{3}}y^{\frac{5}{6}} + x^{\frac{5}{2}}y^{\frac{5}{3}} + x^{\frac{5}{3}}y^{\frac{5}{2}} + x^{\frac{5}{6}}y^{\frac{10}{3}} + y^{\frac{25}{6}})}{x^4 + x^3y + x^2y^2 + xy^3 + y^4} \geqslant 0.$$

$\square$

How do we come up with this? Just recall how we prove the AM-GM inequality.

So far we have not done much expansion. Let us try the following.

Problem 1.6. Let a, b, c be positive real numbers. Prove that

$$\frac{b^2c^2}{a(b+c)} + \frac{c^2a^2}{b(c+a)} + \frac{a^2b^2}{c(a+b)} \geqslant \frac{ab+bc+ca}{2} + \frac{a(b-c)^2}{4(b+c)} + \frac{b(c-a)^2}{4(c+a)} + \frac{c(a-b)^2}{4(a+b)}.$$

Solution. We have

$$\begin{aligned}
& \sum_{\text{cyc}} \frac{b^2c^2}{a(b+c)} \geqslant \frac{1}{2}\sum_{\text{cyc}} ab + \sum_{\text{cyc}} \frac{a(b-c)^2}{4(b+c)} \\
\Leftrightarrow\quad & 4\sum_{\text{cyc}} b^3c^3(a+b)(a+c) \geqslant 2abc\Big(\sum_{\text{cyc}} ab\Big)\prod_{\text{cyc}}(a+b) + abc\sum_{\text{cyc}} a(b-c)^2(a+b)(a+c) \\
\Leftrightarrow\quad & 4\sum_{\text{cyc}} a^4b^4 \geqslant 4\sum_{\text{cyc}} a^4b^2c^2 \\
\Leftrightarrow\quad & 2\sum_{\text{cyc}} a^4(b^2-c^2)^2 \geqslant 0.
\end{aligned}$$

This clearly holds. □

The inequality in the above problem is a stronger one than IMO 1995 Problem 2: Let a, b, c be positive real numbers such that $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geqslant \frac{3}{2}.$$

The above problem asserts that the left-hand side is at least $\dfrac{ab+bc+ca}{2}$, which is at least $\dfrac{3}{2}$ by using $abc = 1$ and the AM-GM inequality (or factorization!).

Here is another old IMO problem that can be solved by completing square.

Problem 1.7. (IMO Shortlist 1999 A1/Problem 2) Let n be a fixed integer, with $n \geqslant 2$.

(a) Determine the least constant C such that the inequality

$$\sum_{1\leqslant i<j\leqslant n} x_ix_j(x_i^2 + x_j^2) \leqslant C\Big(\sum_{1\leqslant i\leqslant n} x_i\Big)^4$$

holds for all real numbers $x_1, \ldots, x_n \geqslant 0$.

(b) For this constant C, determine when equality holds.

Solution. The answer is $C = \frac{1}{8}$. We have

$$\begin{aligned}
&\frac{1}{8}\Big(\sum_{1\leqslant i\leqslant n} x_i\Big)^4 - \sum_{1\leqslant i<j\leqslant n} x_i x_j (x_i^2 + x_j^2) \\
&= \frac{1}{8}\Big(\sum_{1\leqslant i\leqslant n} x_i^2 + 2\sum_{1\leqslant i<j\leqslant n} x_i x_j\Big)^2 - \Big(\sum_{1\leqslant i\leqslant n} x_i^2\Big)\Big(\sum_{1\leqslant i<j\leqslant n} x_i x_j\Big) + \sum_{\substack{1\leqslant i<j\leqslant n\\ 1\leqslant k\leqslant n\\ k\neq i,j}} x_i x_j x_k^2 \\
&= \frac{1}{8}\Big(\sum_{1\leqslant i\leqslant n} x_i^2 - 2\sum_{1\leqslant i<j\leqslant n} x_i x_j\Big)^2 + \sum_{\substack{1\leqslant i<j\leqslant n\\ 1\leqslant k\leqslant n\\ k\neq i,j}} x_i x_j x_k^2 \\
&\geqslant 0.
\end{aligned}$$

Equality holds when two of $x_1, x_2, \ldots, x_n$ are the same and the others are zero. □

Is there any example from more recent competitions? Yes, of course!

Problem 1.8. (APMO 2022 Problem 5) Let a, b, c, d be real numbers such that $a^2 + b^2 + c^2 + d^2 = 1$. Determine the minimum value of $(a-b)(b-c)(c-d)(d-a)$ and determine all values of (a, b, c, d) such that the minimum value is achieved.

Solution. Without loss of generality(WLOG) assume $a \geqslant b, c, d$ and $b \geqslant d$ (since we can swap b and d).

If $c \geqslant b$, then

$$(a-b)(b-c)(c-d)(d-a) = (a-b)(c-b)(c-d)(a-d) \geqslant 0.$$

If $b > c$, then

$$\begin{aligned}
&(a-b)(b-c)(c-d)(d-a) \\
&= -\frac{1}{32}\big(2(a^2+b^2+c^2+d^2) - (a+d)^2 - (b+c)^2\big)^2 \\
&\quad + \frac{1}{32}\big((a-b+c-d)^2 + 2(b-c)(d-a)\big)^2 + \frac{1}{4}(a-b-c+d)^2(b-c)(a-d) \\
&\geqslant -\frac{1}{32}\big(2(a^2+b^2+c^2+d^2) - (a+d)^2 - (b+c)^2\big)^2 \\
&\geqslant -\frac{1}{32}\big(2(a^2+b^2+c^2+d^2)\big)^2 = -\frac{1}{8}.
\end{aligned}$$

So it is clear that $(a-b)(b-c)(c-d)(d-a) \geqslant -\frac{1}{8}$ in any case. Equality holds when $d = -a$, $c = -b$ and $a^2 - 4ab + b^2 = 0$. Together with $a^2 + b^2 + c^2 + d^2 = 1$, it is not hard to see that equality holds when

$$(a, b, c, d) = \left(\frac{\sqrt{3}+1}{4}, \frac{\sqrt{3}-1}{4}, \frac{1-\sqrt{3}}{4}, -\frac{\sqrt{3}+1}{4}\right), \left(\frac{\sqrt{3}+1}{4}, -\frac{\sqrt{3}+1}{4}, \frac{1-\sqrt{3}}{4}, \frac{\sqrt{3}-1}{4}\right)$$

or their cyclic permutations. □

2 Muirhead's Theorem and Schur's Inequality

Although completing square can be used to prove many inequalities, sometimes it is inconvenient to rewrite everything as a sum of squares, and such solutions may be too artificial. Therefore, we should explore more tools that allow us to prove inequalities via expansion. The most useful results which can prove some polynomial inequalities (mainly in 3 variables) are Muirhead's theorem and Schur's inequality. We merely state the two results below. For more details, please see **Inequalities 5.0**.

Theorem 2.1. (Muirhead's theorem(米爾黑德定理)) Let $n \geqslant 2$. For any positive real numbers $a_1, a_2, \dots, a_n$ and nonnegative real numbers $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$ satisfying $(x_1, x_2, \dots, x_n) \succ (y_1, y_2, \dots, y_n)$, we have

$$\sum_{\text{sym}} a_1^{x_1} a_2^{x_2} \cdots a_n^{x_n} \geqslant \sum_{\text{sym}} a_1^{y_1} a_2^{y_2} \cdots a_n^{y_n}.$$

Equality holds if and only if(iff) $a_1 = a_2 = \cdots = a_n$ or $(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_n)$.

Theorem 2.2. (Schur's inequality(舒爾不等式)) For any nonnegative real numbers a, b, c and $r > 0$, we have

$$a^r(a-b)(a-c) + b^r(b-c)(b-a) + c^r(c-a)(c-b) \geqslant 0.$$

Equality holds iff $a = b = c$, or one of a, b, c is zero while the other two are equal.

There are tonnes of classical inequalities that can be proved via these two theorems. We randomly select some examples below.

Problem 2.1. (IMO HK TST 2004 Test 1 Problem 2) Let a, b, c be positive real numbers. Prove that $\dfrac{a^4}{bc} + \dfrac{b^4}{ca} + \dfrac{c^4}{ab} \geqslant a^2 + b^2 + c^2$.

Solution. Multiplying both sides by abc, the inequality is the same as

$$\frac{1}{2} \sum_{\text{sym}} a^5 \geqslant \frac{1}{2} \sum_{\text{sym}} a^3 bc.$$

This holds by Muirhead's theorem since $(5, 0, 0) \succ (3, 1, 1)$. □

Problem 2.2. Let a, b, c be positive real numbers. Prove that

$$\frac{a+b}{a^2+b^2} + \frac{b+c}{b^2+c^2} + \frac{c+a}{c^2+a^2} \geqslant \frac{3(a^3+b^3+c^3)}{a^4+b^4+c^4}.$$

Solution. Multiplying both sides by $(a^2+b^2)(b^2+c^2)(c^2+a^2)(a^4+b^4+c^4)$ and expanding everything, we see that the inequality is the same as

$$\sum_{\text{sym}}\left((a^8b-a^7b^2)+2(a^6b^3-a^5b^4)+2(a^6b^2c-a^5b^2c^2)+(a^4b^4c-a^4b^3c^2)\right)\geqslant 0.$$

This holds by Muirhead's theorem since

$$\begin{aligned}(8,1,0)&\succ(7,2,0),\\(6,3,0)&\succ(5,4,0),\\(6,2,1)&\succ(5,2,2),\\(4,4,1)&\succ(4,3,2).\end{aligned}$$

□

Problem 2.3. Let a,b,c be positive real numbers. Prove that

$$\frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}+\frac{4abc}{(a+b)(b+c)(c+a)}\geqslant 2.$$

Solution. Multiplying both sides by $(a+b)(b+c)(c+a)$ and expanding everything, we see that the inequality is the same as

$$a^3+b^3+c^3+3abc\geqslant a^2b+a^2c+b^2a+b^2c+c^2a+c^2b.$$

This is exactly Schur's inequality. □

Sometimes we are given conditions like $a+b+c=1$ and $abc=1$. We can often use these conditions to homogenize the inequalities so that Muirhead's theorem and Schur's inequality can be used.

Problem 2.4. (CWMO 2005 Problem 7) Let a,b,c be positive real numbers satisfying $a+b+c=1$. Prove that

$$10(a^3+b^3+c^3)-9(a^5+b^5+c^5)\geqslant 1.$$

Solution. Homogenizing the inequality, we see that

$$\begin{aligned}&5\sum_{\text{sym}}a^3-\frac{9}{2}\sum_{\text{sym}}a^5\geqslant 1\\ \Leftrightarrow\quad&5(a+b+c)^2\sum_{\text{sym}}a^3-\frac{9}{2}\sum_{\text{sym}}a^5\geqslant(a+b+c)^5\\ \Leftrightarrow\quad&15\sum_{\text{sym}}a^4b\geqslant 15\sum_{\text{sym}}a^2b^2c.\end{aligned}$$

This holds by Muirhead's theorem since $(4,1,0)\succ(2,2,1)$. □

Problem 2.5. (Tournament of Towns 1997 Spring Senior A Level Problem 5) Let a, b, c be positive numbers such that $abc = 1$. Prove that

$$\frac{1}{a+b+1} + \frac{1}{b+c+1} + \frac{1}{c+a+1} \leqslant 1.$$

Solution. Homogenizing the inequality, we need to prove

$$\frac{1}{a+b+\sqrt[3]{abc}} + \frac{1}{b+c+\sqrt[3]{abc}} + \frac{1}{c+a+\sqrt[3]{abc}} \leqslant \frac{1}{\sqrt[3]{abc}}.$$

In order to avoid handling radicals, we let $x = \sqrt[3]{a}$, $y = \sqrt[3]{b}$ and $z = \sqrt[3]{c}$. Then this becomes

$$\frac{1}{x^3+y^3+xyz} + \frac{1}{y^3+z^3+xyz} + \frac{1}{z^3+x^3+xyz} \leqslant \frac{1}{xyz}.$$

Clearing denominators, this is the same as

$$\sum_{\text{sym}} x^6y^3 \geqslant \sum_{\text{sym}} x^5y^2z^2.$$

This holds by Muirhead's theorem since $(6,3,0) \succ (5,2,2)$. □

Problem 2.6. Let a, b, c be positive real numbers satisfying $abc = 1$. Prove that

$$\frac{a}{c} + \frac{b}{a} + \frac{c}{b} + 3 \geqslant ab + bc + ca + a + b + c.$$

Solution. Since the inequality is cyclic, we use the substitution

$$a = \frac{x}{y},\ b = \frac{y}{z},\ c = \frac{z}{x}.$$

The inequality becomes

$$\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} + 3 \geqslant \frac{x}{z} + \frac{y}{x} + \frac{z}{y} + \frac{x}{y} + \frac{y}{z} + \frac{z}{x}.$$

Multiplying both sides by xyz, this is the same as

$$x^3 + y^3 + z^3 + 3xyz \geqslant x^2y + x^2z + y^2x + y^2z + z^2x + z^2y.$$

This is exactly Schur's inequality. □

Problem 2.7. (IMO Shortlist 2005 A5/Problem 3) Let x, y, z be three positive reals such that $xyz \geqslant 1$. Prove that

$$\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{x^2 + y^5 + z^2} + \frac{z^5 - z^2}{x^2 + y^2 + z^5} \geqslant 0.$$

Solution. Let $xyz = k \geqslant 1$. Homogenizing the inequality, we need to prove

$$\sum_{\text{cyc}} \frac{kx^5 - x^3yz}{kx^5 + xy^3z + xyz^3} \geqslant 0.$$

Clearing denominators, this means

$$\sum_{\text{sym}} \left(\frac{k}{2}x^{11}y^2z^2 - \frac{1}{2}x^9y^3z^3 + 2k^2x^8y^6z - kx^7y^6z^2 - x^7y^5z^3 \right.$$
$$\left. + \frac{k}{2}x^7y^4z^4 - \frac{k^2}{2}x^6y^6z^3 + \frac{k^3-1}{2}x^5y^5z^5 \right) \geqslant 0.$$

Firstly, we have

$$\sum_{\text{sym}} \left(\frac{k^2}{2}x^8y^6z + \frac{k}{2}x^7y^4z^4 \right) \geqslant \sum_{\text{sym}} \left(\frac{1}{2}x^8y^6z + \frac{1}{2}x^7y^4z^4 \right) \geqslant \sum_{\text{sym}} x^{7.5}y^5z^{2.5} \tag{1}$$

by the AM-GM inequality. Then it remains to use the bound $k \geqslant 1$ and Muirhead's theorem to deduce

$$\sum_{\text{sym}} \frac{k}{2}x^{11}y^2z^2 \geqslant \sum_{\text{sym}} \frac{1}{2}x^{11}y^2z^2 \geqslant \sum_{\text{sym}} \frac{1}{2}x^9y^3z^3, \tag{2}$$

$$\sum_{\text{sym}} k^2x^8y^6z \geqslant \sum_{\text{sym}} kx^8y^6z \geqslant \sum_{\text{sym}} kx^7y^6z^2, \tag{3}$$

$$\sum_{\text{sym}} \frac{k^2}{2}x^8y^6z \geqslant \sum_{\text{sym}} \frac{k^2}{2}x^6y^6z^3, \tag{4}$$

$$\sum_{\text{sym}} x^{7.5}y^5z^{2.5} \geqslant \sum_{\text{sym}} x^7y^5z^3, \tag{5}$$

$$\sum_{\text{sym}} \frac{k^3-1}{2}x^5y^5z^5 \geqslant 0. \tag{6}$$

Adding all inequalities from (1) to (6), we obtain the desired inequality. $\square$

How about inequalities with more than 3 variables? Muirhead's theorem can still be applied.

Problem 2.8. (CHKMO 2005 Problem 3) Let a, b, c, d be positive real numbers such that $a + b + c + d = 1$. Prove that

$$6(a^3 + b^3 + c^3 + d^3) \geqslant (a^2 + b^2 + c^2 + d^2) + \frac{1}{8}.$$

Solution. We have

$$\sum_{\text{sym}} a^3 \geqslant \frac{1}{6} \sum_{\text{sym}} a^2 + \frac{1}{8}$$

$$\Leftrightarrow \quad \sum_{\text{sym}} a^3 \geqslant \frac{1}{6}(a+b+c+d) \sum_{\text{sym}} a^2 + \frac{1}{8}(a+b+c+d)^3$$

$$\Leftrightarrow \quad \frac{13}{16} \sum_{\text{sym}} a^3 \geqslant \frac{11}{16} \sum_{\text{sym}} a^2b + \frac{1}{8} \sum_{\text{sym}} abc.$$

This holds by Muirhead's theorem since $(3,0,0,0)$ majorizes both $(2,1,0,0)$ and $(1,1,1,0)$. $\square$

Since Muirhead's theorem and Schur's inequality are too powerful in proving symmetric polynomial inequalities, nowadays we normally do not have such problems in the Olympiad.

3 Substitutions

In this section, we are going to introduce a powerful substitution. This can be used if neither Muirhead's theorem nor Schur's inequality is strong enough to prove the inequality. But, of course, one needs to spend much more effort in doing expansion after using this substitution. Before we introduce this substitution, we first go through an example to explain the idea behind the substitution.

Problem 3.1. (CGMO 2009 Problem 5) Let x, y, z be real numbers greater than or equal to 1. Prove that

$$(x^2 - 2x + 2)(y^2 - 2y + 2)(z^2 - 2z + 2) \leqslant (xyz)^2 - 2xyz + 2.$$

Solution. Let $a = x - 1$, $b = y - 1$ and $c = z - 1$. In other words, $x = a + 1$, $y = b + 1$ and $z = c + 1$. Then $a, b, c \geqslant 0$ and

$$\begin{aligned} & (x^2 - 2x + 2)(y^2 - 2y + 2)(z^2 - 2z + 2) \leqslant (xyz)^2 - 2xyz + 2 \\ \Leftrightarrow \quad & (a^2 + 1)(b^2 + 1)(c^2 + 1) \leqslant \big((a + 1)(b + 1)(c + 1) - 1\big)^2 + 1 \\ \Leftrightarrow \quad & 0 \leqslant \sum_{\text{sym}} (a^2b^2c + 2a^2bc + 2a^2b + ab + abc). \end{aligned}$$

This is obviously true. □

The original problem looks far from trivial. But after a simple substitution, everything becomes clear. This is not that surprising because we can easily observe 0 is a root to a polynomial equation while a little job is needed to verify that some nonzero real number (even an integer) is a root. Therefore, making 0 an equality case can be useful.

Note that most symmetric inequalities have equality $a = b = c$. By assuming $a \geqslant b \geqslant c$ (which can be done if the expression is symmetric), we may consider the substitution $b = c + t$ and $a = c + t + u$, where $t, u \geqslant 0$. After this substitution, the inequality usually turns into an obvious one, as we shall see.

Problem 3.2. (Schur's inequality) Let a, b, c be nonnegative real numbers. Prove that

$$a(a - b)(a - c) + b(b - c)(b - a) + c(c - a)(c - b) \geqslant 0.$$

Solution. WLOG assume $a \geqslant b \geqslant c$. Let $b = c + t$ and $a = c + t + u$ where $t, u \geqslant 0$. Then

$$\begin{aligned} & a(a - b)(a - c) + b(b - c)(b - a) + c(c - a)(c - b) \geqslant 0 \\ \Leftrightarrow \quad & (c + t + u)u(t + u) + (c + t)t(-u) + c(-t - u)(-t) \geqslant 0 \\ \Leftrightarrow \quad & (t^2 + tu + u^2)c + (2tu^2 + u^3) \geqslant 0. \end{aligned}$$

This is obviously true as $c, t, u \geqslant 0$. □

The above proof is essentially the well-known proof of Schur's inequality, but rewritten as a clearer form. Also, we can easily deduce the equality cases based on this proof.

When expanding the expressions using this substitution, we cannot use the symmetric or the cyclic sums, since c, t, u clearly play different roles. It is a usual practice to write everything as a polynomial in c, and for each coefficient (in t, u), we shall write in accordance to the degrees. Note that a homogeneous inequality in a, b, c will be transformed to a homogeneous inequality in c, t, u, so there is still in general a pattern to do the expansion.

Problem 3.3. Let a, b, c be nonnegative real numbers. Prove that

$$\sum_{\text{sym}} (a^4 - 6a^3b + 8a^2b^2 - 3a^2bc) \geqslant 0.$$

Solution. WLOG assume $a \geqslant b \geqslant c$. Let $b = c + t$ and $a = c + t + u$ where $t, u \geqslant 0$. After a tedious expansion, we see that the inequality becomes

$$(2t^2 + 2tu + 2u^2)c^2 + (8t^3 + 12t^2u - 4tu^2 - 4u^3)c + (8t^4 + 16t^3u + 10t^2u^2 + 2tu^3 + 2u^4) \geqslant 0,$$

i.e.

$$2tu(c-u)^2 + 2u^2(c-u)^2 + 2t^2c^2 + (8t^3 + 12t^2u)c + (8t^4 + 16t^3u + 10t^2u^2) \geqslant 0.$$

This is clearly true. Equality holds iff $a = b = c$, or $a = 2b = 2c$ or its permutation. □

The same method can be used to solve inequalities with cyclic expressions. A better way is to consider $a \geqslant b \geqslant c$ and $a \leqslant b \leqslant c$ separately.

Problem 3.4. Let a, b, c be nonnegative real numbers satisfying $a^2 + b^2 + c^2 = 1$. Prove that

$$(a-b)(b-c)(c-a)(a+b+c) \leqslant \frac{1}{4}.$$

Solution. The inequality is trivial when $a \geqslant b \geqslant c$, since the left-hand side is non-positive. For $a \leqslant b \leqslant c$, we shall prove that

$$(a-b)(b-c)(c-a)(a+b+c) \leqslant \frac{1}{4}(a^2 + b^2 + c^2)^2.$$

Let $b = a + t$ and $c = a + t + u$ where $t, u \geqslant 0$. Then this inequality is equivalent to

$$4(-t)(-u)(t+u)(3a + 2t + u) \leqslant \left(a^2 + (a+t)^2 + (a+t+u)^2\right)^2.$$

After expansion, we see that this becomes

$$9a^4 + (24t + 12u)a^3 + (28t^2 + 28tu + 10u^2)a^2 + (16t^3 + 12t^2u + 4tu^2 + 4u^3)a + (2t^2 - u^2)^2 \geqslant 0.$$

This is clearly true. Equality holds iff $(a,b,c)=\left(0,\dfrac{\sqrt{2-\sqrt{2}}}{2},\dfrac{(1+\sqrt{2})\sqrt{2-\sqrt{2}}}{2}\right)$ or its cyclic permutation. $\square$

The above equality case is indeed $\left(0,\sin\dfrac{\pi}{8},\cos\dfrac{\pi}{8}\right)$. It is very difficult to find such an equality case in general. But this substitution turns the problem into a trivial one.

The same idea also works if there are more than 3 variables. For a cyclic inequality in 4 variables, it is common that the equality holds when $a=c$ and $b=d$. Therefore, we may consider the substitution $a=c+t$ and $b=d+u$.

Problem 3.5. (IMO Shortlist 2008 A7) Prove that for any four positive real numbers a,b,c,d the inequality

$$\frac{(a-b)(a-c)}{a+b+c}+\frac{(b-c)(b-d)}{b+c+d}+\frac{(c-d)(c-a)}{c+d+a}+\frac{(d-a)(d-b)}{d+a+b}\geqslant 0$$

holds. Determine all cases of equality.

Solution. WLOG assume $a\geqslant c$ and $b\geqslant d$ (this can be done if the expression is cyclic). Let $a=c+t$ and $b=d+u$ where $t,u\geqslant 0$. The inequality is the same as

$$\frac{(c-d+t-u)t}{2c+d+t+u}+\frac{(-c+d+u)u}{c+2d+u}+\frac{-(c-d)t}{2c+d+t}+\frac{-(-c+d-t)u}{c+2d+t+u}\geqslant 0.$$

Clearing denominators, this is the same as

$$\begin{aligned}\left(2\left(t-\frac{3}{4}u\right)^2+\frac{23}{8}u^2\right)c^3+(9t^2+12u^2)c^2d+(12t^2+9u^2)cd^2+(4t^2+3tu+2u^2)d^3\\ +(3t^3+6tu^2+6u^3)c^2+(9t^3+12t^2u+12tu^2+9u^3)cd\\ +(6t^3+6t^2u+9tu^2+3u^3)d^2+(t^4+3t^3u+6t^2u^2+6tu^3+2u^4)c\\ +(2t^4+6t^3u+6t^2u^2+6tu^3+u^4)d+(t^4u+2t^3u^2+2t^2u^3+tu^4)\geqslant 0.\end{aligned}$$

This clearly holds. Equality holds when $a=c$ and $b=d$. $\square$

If this is not shocking, let us go through one more problem. One should really try to do the expansion by hand!

Problem 3.6. (IMO Shortlist 2010 A8) Given six positive numbers a,b,c,d,e,f such that $a<b<c<d<e<f$. Let $a+c+e=S$ and $b+d+f=T$. Prove that

$$2ST>\sqrt{3(S+T)\big(S(bd+bf+df)+T(ac+ae+ce)\big)}.$$

Solution. Let $a = u$, $b = u + v$, $c = u + v + w$, $d = u + v + w + x$, $e = u + v + w + x + y$ and $f = u + v + w + x + y + z$ where $u, v, w, x, y, z > 0$. Squaring both sides of the desired inequality and expanding everything, we see that the inequality is equivalent to

$$
\begin{aligned}
&(36vw + 18vy + 36w^2 + 54wx + 36wy + 18wz + 27x^2 + 54xy + 36y^2 + 36yz)u^2 \\
&+ (78v^2w + 39v^2y + 108vw^2 + 129vwx + 108vwy + 21vwz + 48vx^2 + 114vxy + 72vy^2 \\
&+ 51vyz + 48w^3 + 108w^2x + 72w^2y + 36w^2z + 87wx^2 + 144wxy + 30wxz + 72wy^2 \\
&+ 72wyz + 15wz^2 + 27x^3 + 75x^2y + 6x^2z + 72xy^2 + 51xyz + 24y^3 + 36y^2z + 12yz^2)u \\
&+ (42w + 21y)v^3 + \left(73w^2 + 83wx + 73wy + 10wz + \frac{87}{4}x^2 + 64xy + 37y^2 + 20yz\right)v^2 \\
&+ \Bigg(56w^3 + 116w^2x + 84w^2y + 32w^2z + 85wx^2 + 149wxy + 21wxz + 72wy^2 + 59wyz \\
&+ 8wz^2 + 25x^3 + 71x^2y + \frac{13}{2}x^2z + 68xy^2 + 39xyz + 22y^3 + 29y^2z + 7yz^2\Bigg)v \\
&+ 16w^4 + (48x + 32y + 16z)w^3 + (55x^2 + 84xy + 26xz + 36y^2 + 36yz + 7z^2)w^2 \\
&+ (30x^3 + 76x^2y + 14x^2z + 66xy^2 + 50xyz + 4xz^2 + 20y^3 + 30y^2z + 10yz^2)w \\
&+ 7x^4 + (24y + 4z)x^3 + \left(31y^2 + 19yz + \frac{3}{4}z^2\right)x^2 + (18y^3 + 23y^2z + 5yz^2)x \\
&+ 4y^4 + 8y^3z + 4y^2z^2 + \left(3(v - z)u + v(3v - z) + \frac{1}{2}x(5v - z)\right)^2 > 0
\end{aligned}
$$

This is obviously true. $\square$

4 Degree 4 Polynomial Inequalities

In this section, we state two results regarding degree 4 polynomial inequalities. Note that these are not formal theorems so the ways to derive such results are more important.

Proposition 4.1. Let α, β, γ be real numbers satisfying $\gamma \geqslant \dfrac{\alpha^2 + \alpha\beta + \beta^2}{3} - 1$. Then

$$\sum_{\text{cyc}} \left(a^4 + \alpha a^3 b + \beta a b^3 + \gamma a^2 b^2 - (1 + \alpha + \beta + \gamma) a^2 bc\right) \geqslant 0$$

for any real numbers a, b, c.

Proof. We use the following identities.

$$\begin{aligned}
\sum_{\text{cyc}} (a^4 - a^2 b^2) &= \frac{1}{6} \sum_{\text{cyc}} (a^2 + b^2 - 2c^2)^2, \\
\sum_{\text{cyc}} (a^3 b - a^2 bc) &= \frac{1}{3} \sum_{\text{cyc}} (ab - ac)(a^2 + b^2 - 2c^2), \\
\sum_{\text{cyc}} (ab^3 - a^2 bc) &= \frac{1}{3} \sum_{\text{cyc}} (ab - bc)(a^2 + b^2 - 2c^2), \\
\sum_{\text{cyc}} (a^2 b^2 - a^2 bc) &= \frac{1}{2(\alpha^2 + \alpha\beta + \beta^2)} \sum_{\text{cyc}} \left((\alpha + \beta)ab - \alpha ac - \beta bc\right)^2.
\end{aligned}$$

Then

$$\begin{aligned}
&\sum_{\text{cyc}} \left(a^4 + \alpha a^3 b + \beta a b^3 + \gamma a^2 b^2 - (1 + \alpha + \beta + \gamma) a^2 bc\right) \\
&= \sum_{\text{cyc}} (a^4 - a^2 b^2) + \alpha \sum_{\text{cyc}} (a^3 b - a^2 bc) + \beta \sum_{\text{cyc}} (ab^3 - a^2 bc) + (\gamma + 1) \sum_{\text{cyc}} (a^2 b^2 - a^2 bc) \\
&= \frac{1}{6} \sum_{\text{cyc}} (a^2 + b^2 - 2c^2)^2 + \frac{\alpha}{3} \sum_{\text{cyc}} (ab - ac)(a^2 + b^2 - 2c^2) + \frac{\beta}{3} \sum_{\text{cyc}} (ab - bc)(a^2 + b^2 - 2c^2) \\
&\quad + \frac{\gamma + 1}{2(\alpha^2 + \alpha\beta + \beta^2)} \sum_{\text{cyc}} \left((\alpha + \beta)ab - \alpha ac - \beta bc\right)^2 \\
&= \frac{1}{6} \sum_{\text{cyc}} \left((a^2 + b^2 - 2c^2) + \alpha(ab - ac) + \beta(ab - bc)\right)^2 \\
&\quad + \left(\frac{\gamma + 1}{2(\alpha^2 + \alpha\beta + \beta^2)} - \frac{1}{6}\right) \sum_{\text{cyc}} \left((\alpha + \beta)ab - \alpha ac - \beta bc\right)^2 \\
&\geqslant 0
\end{aligned}$$

since $\gamma \geqslant \dfrac{\alpha^2 + \alpha\beta + \beta^2}{3} - 1$. $\square$

Note that the above result does not provide any information if $\gamma < \dfrac{\alpha^2+\alpha\beta+\beta^2}{3} - 1$. So it is possible that there is a stronger result. An example of an application of proposition 4.1 is problem 1.2, where we have to prove

$$(a^2+b^2+c^2)^2 \geqslant 3(a^3b+b^3c+c^3a)$$

for any real numbers a, b, c. Indeed, we have $\alpha = -3$, $\beta = 0$ and $\gamma = 2$ in this problem. The relation $\gamma \geqslant \dfrac{\alpha^2+\alpha\beta+\beta^2}{3} - 1$ is satisfied, so proposition 4.1 can be used.

Proposition 4.2. Let a, b, c be nonnegative real numbers. Then

$$\sum_{\text{sym}} \left(a^4 + \alpha a^3 b + \beta a^2 b^2 - (1+\alpha+\beta)a^2bc\right) \geqslant 0$$

iff one of the following holds:

- $\alpha \geqslant -2$ and $\beta \geqslant -\alpha - 2$; or
- $\alpha < -2$ and $\beta \geqslant \dfrac{1}{4}\alpha^2 - 1$.

Proof. WLOG assume $a \geqslant b \geqslant c$. Let $b = c + t$ and $a = c + t + u$ where $t, u \geqslant 0$. After expansion, the inequality is equivalent to

$$\begin{aligned} &(4\alpha+2\beta+10)(t^2+tu+u^2)c^2 \\ &+\left((6\alpha+4\beta+12)t^3+(9\alpha+6\beta+18)t^2u+(7\alpha+2\beta+22)tu^2+(2\alpha+8)u^3\right)c \\ &+\left((2\alpha+2\beta+4)t^4+(4\alpha+4\beta+8)t^3u+(3\alpha+2\beta+12)t^2u^2+(\alpha+8)tu^3+2u^4\right) \geqslant 0. \end{aligned}$$

There are three separated cases to consider.

Case 1. $\alpha \geqslant -2$

When $(c, t, u) = (0, 1, 0)$, we need $2\alpha + 2\beta + 4 \geqslant 0$, i.e. $\beta \geqslant -\alpha - 2$. We claim that this is also a sufficient condition. Note that the inequality is linear in β with positive coefficients. So it suffices to check the case $\beta = -\alpha - 2$. This is the same as

$$\begin{aligned} (2\alpha+6)(t^2+tu+u^2)c^2 + \left((2\alpha+4)t^3+(3\alpha+6)t^2u+(5\alpha+18)tu^2+(2\alpha+8)u^3\right)c \\ +\left((\alpha+8)t^2u^2+(\alpha+8)tu^3+2u^4\right) \geqslant 0. \end{aligned}$$

This is true as each coefficient is nonnegative from the condition $\alpha \geqslant -2$. For $\alpha \geqslant -2$, the inequality is true iff $\beta \geqslant -\alpha - 2$. When $\beta = -\alpha - 2$, equality holds iff $a = b = c$, or $a = b$ and $c = 0$ or its permutation.

Case 2. $-2 > \alpha \geqslant -4$

When $(t, u) = (1, 0)$, we need $(2\alpha+\beta+5)c^2 + (3\alpha+2\beta+6)c + (\alpha+\beta+2) \geqslant 0$. As it is nonnegative for all c, its leading coefficient must be nonnegative.

If $2\alpha+\beta+5=0$, then $(-\alpha-4)c+(-\alpha-3)\geqslant 0$. Note that $\alpha=-4$ implies $\beta=3$, which satisfies $\beta\geqslant\frac{1}{4}\alpha^2-1$. Otherwise, $c\leqslant-\frac{\alpha+3}{\alpha+4}$, which is not true for large c.

If $(2\alpha+\beta+5)c^2+(3\alpha+2\beta+6)c+(\alpha+\beta+2)$ has positive leading coefficient, Its minimum value in the domain $c\in[0,\infty)$ is attained at $c=0$ or at the axis of symmetry. In the former case, we need $3\alpha+2\beta+6\geqslant 0$ (which implies $c<0$ at the axis of symmetry), while in the latter case, we need $\Delta\leqslant 0$. Now,

$$\begin{aligned}
&\Delta=(3\alpha+2\beta+6)^2-4(2\alpha+\beta+5)(\alpha+\beta+2)\leqslant 0\\
\Leftrightarrow\quad&\alpha^2-4\beta-4\leqslant 0\\
\Leftrightarrow\quad&\beta\geqslant\frac{1}{4}\alpha^2-1.
\end{aligned}$$

Also, we observe that $3\alpha+2\beta+6\geqslant 0$ implies $\beta\geqslant\frac{1}{4}\alpha^2-1$. Indeed, it suffices to show $-\frac{3}{2}\alpha-3\geqslant\frac{1}{4}\alpha^2-1$. This is the same as $\frac{1}{4}(\alpha+2)(\alpha+4)\leqslant 0$, which is true from $-2>\alpha\geqslant-4$. Therefore, we always have $\beta\geqslant\frac{1}{4}\alpha^2-1$. Conversely, we shall check that the inequality is true in the case $\beta=\frac{1}{4}\alpha^2-1$. We find that

$$\begin{aligned}
&\left(\frac{1}{2}\alpha^2+4\alpha+8\right)(t^2+tu+u^2)c^2\\
&\left((\alpha^2+6\alpha+8)t^3+\left(\frac{3}{2}\alpha^2+9\alpha+12\right)t^2u+\left(\frac{1}{2}\alpha^2+7\alpha+20\right)tu^2+(2\alpha+8)u^3\right)c\\
&\left(\left(\frac{1}{2}\alpha^2+2\alpha+2\right)t^4+(\alpha^2+4\alpha+4)t^3u+\left(\frac{1}{2}\alpha^2+3\alpha+10\right)t^2u^2+(\alpha+8)tu^3+2u^4\right)\geqslant 0\\
\Leftrightarrow\quad&\frac{1}{2}\left((\alpha+4)tc+\frac{1}{2}(\alpha+4)uc+(\alpha+2)t^2+(\alpha+2)tu\right)^2+\frac{3}{8}(\alpha+4)^2u^2c^2\\
&+4(\alpha+4)tu^2c+2(\alpha+4)u^3c+(\alpha+8)t^2u^2+(\alpha+8)tu^3+2u^4\geqslant 0.
\end{aligned}$$

This is trivially true from $-2>\alpha\geqslant-4$. For $-2>\alpha\geqslant-4$, the inequality is true iff $\beta\geqslant\frac{1}{4}\alpha^2-1$. When $\beta=\frac{1}{4}\alpha^2-1$, equality holds iff $a=b=c$, or $a=b=-\frac{2}{\alpha+2}c$ or its permutation.

Case 3. $-4>\alpha$

When $(t,u)=(0,1)$, we need $(2\alpha+\beta+5)c^2+(\alpha+4)c+1\geqslant 0$. The leading coefficient should be nonnegative. If $2\alpha+\beta+5=0$, we have $(\alpha+4)c+1\geqslant 0$, or $c\leqslant-\frac{1}{\alpha+4}$, which cannot be true for large c. Suppose $2\alpha+\beta+5>0$. Note that we always have $\alpha+4<0$. Hence, the inequality is true iff

$$\Delta=(\alpha+4)^2-4(2\alpha+\beta+5)\leqslant 0.$$

After expansion, this means $\beta \geqslant \frac{1}{4}\alpha^2 - 1$. Similar to the above cases, it suffices to check for the case $\beta = \frac{1}{4}\alpha^2 - 1$. This is the same as

$$\begin{aligned}
&\left(\frac{1}{2}\alpha^2 + 4\alpha + 8\right)(t^2 + tu + u^2)c^2 \\
&+\left((\alpha^2 + 6\alpha + 8)t^3 + \left(\frac{3}{2}\alpha^2 + 9\alpha + 12\right)t^2u + \left(\frac{1}{2}\alpha^2 + 7\alpha + 20\right)tu^2 + (2\alpha + 8)u^3\right)c \\
&+\left(\left(\frac{1}{2}\alpha^2 + 2\alpha + 2\right)t^4 + (\alpha^2 + 4\alpha + 4)t^3u + \left(\frac{1}{2}\alpha^2 + 3\alpha + 10\right)t^2u^2 + (\alpha + 8)tu^3 + 2u^4\right) \geqslant 0,
\end{aligned}$$

which is equivalent to

$$\begin{aligned}
&\frac{1}{2}\left(\frac{1}{2}(\alpha + 4)tc + (\alpha + 4)uc + \frac{1}{2}(\alpha + 8)tu + 2u^2\right)^2 + \frac{3}{8}(\alpha + 4)^2t^2c^2 \\
&+(\alpha + 2)(\alpha + 4)t^3c + \frac{1}{4}(\alpha + 4)(5\alpha + 4)t^2uc + \frac{1}{2}(\alpha + 2)^2t^4 \\
&+(\alpha + 2)^2t^3u + \frac{1}{8}(2\alpha^2 + (\alpha + 4)^2)t^2u^2 \geqslant 0.
\end{aligned}$$

This is true from $\alpha < -4$. For $\alpha < -4$, the inequality is true iff $\beta \geqslant \frac{1}{4}\alpha^2 - 1$. When $\beta = \frac{1}{4}\alpha^2 - 1$, equality holds iff $a = b = c$ or $-\frac{2}{\alpha + 2}a = b = c$ or its permutation.

$\square$

Indeed, the proof of proposition 4.2 is too complicated to memorize. For our use, we just need to use the substitution as in problem 3.3 (which is also an application of this result). We just state the result here for reference. One can check if the inequality really holds before carrying out the expansion. Note that proposition 4.2 is an equivalent result, so it is the strongest result for degree 4 symmetric polynomial inequalities.

Although both of the above results are about degree 4 inequalities, one may still use them to prove higher degree inequalities as one can multiply it by a nonnegative (symmetric) expression. For example, given a degree 4 polynomial $f(a, b, c)$ with $f(a, b, c) \geqslant 0$, we also have $(a^2 + b^2 + c^2)f(a, b, c) \geqslant 0$ as a degree 6 inequality. If $a, b, c \geqslant 0$, we also have $(a + b + c)f(a, b, c) \geqslant 0$ as a degree 5 inequality, etc.

Problem 4.1. Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 = 1$. Prove that

$$\left(1 - \left(\frac{a + b}{2}\right)^2\right)\left(1 - \left(\frac{b + c}{2}\right)^2\right)\left(1 - \left(\frac{c + a}{2}\right)^2\right) \geqslant \frac{8}{27}.$$

Solution. Homogenizing the inequality and expanding everything, we see that

$$\prod_{\text{cyc}}\left(1-\left(\frac{a+b}{2}\right)^2\right) \geqslant \frac{8}{27}$$

$$\Leftrightarrow \qquad \prod_{\text{cyc}}\left(a^2+b^2+c^2-\left(\frac{a+b}{2}\right)^2\right) \geqslant \frac{8}{27}(a^2+b^2+c^2)^3$$

$$\Leftrightarrow \quad \sum_{\text{sym}}(230a^6-648a^5b+1461a^4b^2-27a^4bc-675a^3b^3-810a^3b^2c+469a^2b^2c^2) \geqslant 0. \tag{1}$$

One may find that both Muirhead's theorem and Schur's inequality are not strong enough to prove this. Thus, we consider

$$(a^2+b^2+c^2-ab-bc-ca)\sum_{\text{sym}}\left(a^4+\alpha a^3b+\beta a^2b^2-(1+\alpha+\beta)a^2bc\right) \tag{2}$$

where α,β satisfy the conditions in proposition 4.2 (note that $a^2+b^2+c^2-ab-bc-ca$ is the strongest nonnegative degree 2 symmetric polynomial in a,b,c). Equation (2) is equivalent to

$$\sum_{\text{sym}}\Big(a^6+(\alpha-2)a^5b+(-\alpha+2\beta+2)a^4b^2+(-2\alpha-\beta-2)a^4bc+(\alpha-\beta)a^3b^3$$
$$-2\beta a^3b^2c+(\alpha+2\beta+1)a^2b^2c^2\Big).$$

The most negative term in (1) is probably $-648a^5b$. So we want to have α as negative as possible. With some trial and error (we usually consider integers α,β), we choose $\alpha=-4$ and $\beta=\frac{1}{4}\alpha^2-1=3$. By proposition 4.2 and $a^2+b^2+c^2-ab-bc-ca \geqslant 0$, we have

$$\sum_{\text{sym}}(a^6-6a^5b+12a^4b^2+3a^4bc-7a^3b^3-6a^3b^2c+3a^2b^2c^2) \geqslant 0. \tag{3}$$

Consider (1)−108×(3). It suffices to prove

$$\sum_{\text{sym}}(122a^6+165a^4b^2-351a^4bc+81a^3b^3-162a^3b^2c+145a^2b^2c^2) \geqslant 0.$$

This is true due to the following inequalities (obtained by the AM-GM inequality or Muirhead's theorem):

$$122\sum_{\text{sym}}(a^6+a^2b^2c^2) = 244\sum_{\text{sym}}\left(\frac{a^6+a^2b^2c^2}{2}\right) \geqslant 244\sum_{\text{sym}}a^4bc,$$
$$107\sum_{\text{sym}}a^4b^2 \geqslant 107\sum_{\text{sym}}a^4bc,$$
$$23\sum_{\text{sym}}(a^4b^2+a^2b^2c^2) = 46\sum_{\text{sym}}\left(\frac{a^4b^2+a^2b^2c^2}{2}\right) \geqslant 46\sum_{\text{sym}}a^3b^2c,$$
$$35\sum_{\text{sym}}a^4b^2 \geqslant 35\sum_{\text{sym}}a^3b^2c,$$
$$81\sum_{\text{sym}}a^3b^3 \geqslant 81\sum_{\text{sym}}a^3b^2c.$$

□

5 The *uvw* Method

This section mainly deals with symmetric inequalities in 3 variables. The key idea of the uvw method is that every symmetric polynomial in three variables a, b, c can be rewritten uniquely as a polynomial in the elementary symmetric polynomials $a+b+c$, $ab+bc+ca$ and abc. Before we state the main result, we first use the same idea in the case of two variables to demonstrate its usefulness.

Problem 5.1. (IMO Shortlist 2008 A2/Problem 2)

(a) Prove that
$$\frac{x^2}{(x-1)^2}+\frac{y^2}{(y-1)^2}+\frac{z^2}{(z-1)^2}\geqslant 1$$
for all real numbers x, y, z, each different from 1, and satisfying $xyz = 1$.

(b) Prove that equality holds above for infinitely many triples of rational numbers x, y, z, each different from 1, and satisfying $xyz = 1$.

Solution. We regard this as an inequality problem in the two variables x and y. Let $t = x + y$. Since it is given that $xy = \frac{1}{z}$, we can express the symmetric polynomials in x, y in terms of t and z. Indeed,

$$\begin{aligned}
& \frac{x^2}{(x-1)^2}+\frac{y^2}{(y-1)^2}+\frac{z^2}{(z-1)^2}\geqslant 1\\
\Leftrightarrow\quad & \frac{x^2(y^2-2y+1)+y^2(x^2-2x+1)}{(x^2-2x+1)(y^2-2y+1)}\geqslant 1-\frac{z^2}{(z-1)^2}\\
\Leftrightarrow\quad & \frac{2x^2y^2-2xy(x+y)+(x^2+y^2)}{x^2y^2-2xy(x+y)+4xy+(x^2+y^2)-2(x+y)+1}\geqslant\frac{(z-1)^2-z^2}{(z-1)^2}\\
\Leftrightarrow\quad & \frac{2\left(\frac{1}{z}\right)^2-\frac{2t}{z}+\left(t^2-\frac{2}{z}\right)}{\left(\frac{1}{z}\right)^2-\frac{2t}{z}+\frac{4}{z}+\left(t^2-\frac{2}{z}\right)-2t+1}\geqslant\frac{-2z+1}{(z-1)^2}\\
\Leftrightarrow\quad & \frac{z^2t^2-2zt+(-2z+2)}{z^2t^2-(2z^2+2z)t+(z^2+2z+1)}\geqslant\frac{-2z+1}{(z-1)^2}.
\end{aligned}$$

This can be viewed as a quadratic polynomial in t, which can be easily factorized. Indeed,

$$\begin{aligned}
& \frac{x^2}{(x-1)^2}+\frac{y^2}{(y-1)^2}+\frac{z^2}{(z-1)^2}\geqslant 1\\
\Leftrightarrow\quad & z^4t^2-(6z^3-2z^2)t+(9z^2-6z+1)\geqslant 0\\
\Leftrightarrow\quad & \left(z^2t-(3z-1)\right)^2\geqslant 0.
\end{aligned}$$

Equality holds iff $tz^2 - 3z + 1 = 0$. Part (b) is not hard in view of this equality case. We omit the proof here. $\square$

In the above solution, we do not use the simple result $t^2 = (x+y)^2 \geqslant 4xy = \dfrac{4}{z}$, though this could be useful in other problems.

Now, we move on to a similar substitution for 3 variables. For any real numbers a, b, c, let

$$\begin{cases} a+b+c = 3u, \\ ab+bc+ca = 3v^2, \\ abc = w^3. \end{cases}$$

This substitution is known as the *uvw method*. The reason for choosing $3u, 3v^2, w^3$ but not simply u, v, w is to rewrite the (probably) equality case $a = b = c$ as a simple relation $u = v = w$. Also, a homogeneous inequality remains homogeneous. Note that we write $ab + bc + ca = 3v^2$ by convention. But this does not mean that $ab + bc + ca$ must be nonnegative. In other words, v can be a non-real number but v^2 must be real. However, this does not really matter as we always deal with even powers of v only.

Analogous to the result $(x + y)^2 \geqslant 4xy$ in the case of two variables, we have an important relationship between the variables u, v, w. This is a strong inequality and is the main component for solving problems when applying the *uvw* substitution.

Theorem 5.1. (*uvw* theorem) For any real numbers a, b, c, denote

$$\begin{cases} a+b+c = 3u, \\ ab+bc+ca = 3v^2, \\ abc = w^3. \end{cases}$$

Then we have

$$3uv^2 - 2u^3 - 2\sqrt{(u^2-v^2)^3} \leqslant w^3 \leqslant 3uv^2 - 2u^3 + 2\sqrt{(u^2-v^2)^3}.$$

Proof. As $a, b, c \in \mathbb{R}$, we have $(a-b)^2(b-c)^2(c-a)^2 \geqslant 0$. Now,

$$\begin{aligned} (a-b)^2(b-c)^2(c-a)^2 &= \Big(\sum_{\text{cyc}} (ab^2 - a^2b)\Big)^2 \\ &= \sum_{\text{sym}} a^4b^2 - 2\sum_{\text{cyc}} a^4bc - 2\sum_{\text{cyc}} a^3b^3 + 2\sum_{\text{sym}} a^3b^2c - 6a^2b^2c^2. \qquad (1) \end{aligned}$$

For simplicity, we first consider a similar expression

$$\Big(\sum_{\text{cyc}} (ab^2 + a^2b)\Big)^2 = \sum_{\text{sym}} a^4b^2 + 2\sum_{\text{cyc}} a^4bc + 2\sum_{\text{cyc}} a^3b^3 + 2\sum_{\text{sym}} a^3b^2c + 6a^2b^2c^2. \qquad (2)$$

Taking the difference of (1) and (2), we get

$$(a-b)^2(b-c)^2(c-a)^2 = \Big(\sum_{\text{cyc}}(ab^2+a^2b)\Big)^2 - 4\sum_{\text{cyc}}a^4bc - 4\sum_{\text{cyc}}a^3b^3 - 12a^2b^2c^2. \tag{3}$$

Now,

$$\begin{aligned}
\Big(\sum_{\text{cyc}}(ab^2+a^2b)\Big)^2 &= \big((a+b+c)(ab+bc+ca)-3abc\big)^2 \\
&= (9uv^2-3w^3)^2 \\
&= 81u^2v^4 - 54uv^2w^3 + 9w^6, \\
\sum_{\text{cyc}}a^4bc &= abc(a^3+b^3+c^3) \\
&= abc\Big((a+b+c)^3 - 3\sum_{\text{sym}}a^2b - 6abc\Big) \\
&= abc\big((a+b+c)^3 - 3\big((a+b+c)(ab+bc+ca)-3abc\big) - 6abc\big) \\
&= w^3(27u^3 - 3(9uv^2-3w^3) - 6w^3) \\
&= -27uv^2w^3 + 27u^3w^3 + 3w^6, \\
\sum_{\text{cyc}}a^3b^3 &= (ab+bc+ca)^3 - 3\sum_{\text{sym}}a^3b^2c - 6a^2b^2c^2 \\
&= 27v^6 - 3w^3(9uv^2-3w^3) - 6w^6 \\
&= 27v^6 - 27uv^2w^3 + 3w^6.
\end{aligned}$$

Putting these into (3), we have

$$\begin{aligned}
0 &\leqslant (a-b)^2(b-c)^2(c-a)^2 \\
&= 81u^2v^4 - 108v^6 + 162uv^2w^3 - 108u^3w^3 - 27w^6 \\
&= 27\big(-w^6 + (6uv^2-4u^3)w^3 + (3u^2v^4-4v^6)\big).
\end{aligned}$$

Viewing this as a quadratic inequality in w^3, we obtain the desired result

$$3uv^2 - 2u^3 - 2\sqrt{(u^2-v^2)^3} \leqslant w^3 \leqslant 3uv^2 - 2u^3 + 2\sqrt{(u^2-v^2)^3}.$$

□

Note that the formula given in the *uvw* theorem is homogeneous. Also w^3 has a unique value iff $u = v$, in which case we must have $u = v = w$. These conditions may help us memorize the formula.

Very often, we deal with inequalities in nonnegative numbers a, b, c. In that case, u, v, w can also be taken as nonnegative real numbers and we have a further result $u \geqslant v \geqslant w$. The first inequality follows from

$$9(u^2-v^2) = (a+b+c)^2 - 3(ab+bc+ca) = \frac{1}{2}\big((a-b)^2+(b-c)^2+(c-a)^2\big) \geqslant 0.$$

The second inequality follows from

$$v^2 = \frac{ab+bc+ca}{3} \geqslant \sqrt[3]{a^2b^2c^2} = w^2$$

by the AM-GM inequality.

Problem 5.2. Let a, b, c be positive real numbers such that $abc = 1$. Prove that

$$\frac{1}{a(1+bc)^2} + \frac{1}{b(1+ca)^2} + \frac{1}{c(1+ab)^2} \leqslant \frac{1}{4} + \frac{4}{(1+ab)(1+bc)(1+ca)}.$$

Solution. It will be too complicated to expand directly. So we use the uvw substitution to help in the expansion. Let

$$\begin{cases} a+b+c = 3u, \\ ab+bc+ca = 3v^2, \\ abc = w^3 \end{cases}$$

where $u, v^2, w^3 \in \mathbb{R}$. Multiplying both sides by $abc(1+ab)^2(1+bc)^2(1+ca)^2$, the left-hand side of the inequality is equal to

$$\begin{aligned} &\sum_{\text{cyc}} ab + 4\sum_{\text{cyc}} a^2bc + \sum_{\text{sym}} a^3b^2c + 12a^2b^2c^2 + 4\sum_{\text{cyc}} a^3b^3c^2 + \sum_{\text{cyc}} a^4b^3c^3 \\ &= 3v^2 + 12u + (9uv^2 - 3) + 12 + 12v^2 + 3u \quad (\text{using } w^3 = abc = 1) \\ &= 9uv^2 + 15u + 15v^2 + 9. \end{aligned} \tag{1}$$

Similarly, the right-hand side of the inequality is equal to

$$\begin{aligned} &abc\left(1 + \sum_{\text{cyc}} ab + \sum_{\text{cyc}} a^2bc + a^2b^2c^2\right)\left(\frac{1}{4}\left(1 + \sum_{\text{cyc}} ab + \sum_{\text{cyc}} a^2bc + a^2b^2c^2\right) + 4\right) \\ &= (3u + 3v^2 + 2)\left(\frac{3u + 3v^2 + 2}{4} + 4\right) \\ &= \frac{9u^2 + 18uv^2 + 60u + 9v^4 + 60v^2 + 36}{4}. \end{aligned} \tag{2}$$

Comparing (1) and (2), we only need to prove

$$36uv^2 + 60u + 60v^2 + 36 \leqslant 9u^2 + 18uv^2 + 60u + 9v^4 + 60v^2 + 36.$$

This is the same as $9(u - v^2)^2 \geqslant 0$, which is obviously true. Equality holds iff $u = v^2$. This means $a + b + c = ab + bc + ca$, or

$$(a-1)(b-1)(c-1) = abc - (ab+bc+ca) + (a+b+c) - 1 = 0.$$

In other words, equality holds iff one of a, b, c is equal to 1. □

In the above problem, we do not use any relations between u, v, w, so it is only a simple usage of the uvw substitution.

Problem 5.3. (IMO Shortlist 2006 A6/Problem 3) Determine the least real number M such that the inequality

$$\left|ab(a^2-b^2)+bc(b^2-c^2)+ca(c^2-a^2)\right| \leqslant M(a^2+b^2+c^2)^2$$

holds for all real numbers a, b and c.

Solution. We have the identity

$$ab(a^2-b^2)+bc(b^2-c^2)+ca(c^2-a^2) = -(a-b)(b-c)(c-a)(a+b+c).$$

It is not easy to handle cyclic expressions as well as terms involving absolute values. So we may square both sides to obtain a symmetric inequality. So we shall prove

$$(a-b)^2(b-c)^2(c-a)^2(a+b+c)^2 \leqslant M^2(a^2+b^2+c^2)^4.$$

Let

$$\begin{cases} a+b+c &= 3u, \\ ab+bc+ca &= 3v^2, \\ abc &= w^3 \end{cases}$$

where $u, v^2, w^3 \in \mathbb{R}$. Recall that

$$(a-b)^2(b-c)^2(c-a)^2 = -27w^6 + (162uv^2 - 108u^3)w^3 + (81u^2v^4 - 108v^6)$$

from the proof of the uvw theorem. So

$$\begin{aligned}
& & (a-b)^2(b-c)^2(c-a)^2(a+b+c)^2 &\leqslant M^2(a^2+b^2+c^2)^4 \\
\Leftrightarrow & & \left(-27w^6 + (162uv^2 - 108u^3)w^3 + (81u^2v^4 - 108v^6)\right)(9u^2) &\leqslant M^2(9u^2-6v^2)^4 \\
\Leftrightarrow & & -3u^2w^6 + (18u^3v^2 - 12u^5)w^3 + (9u^4v^4 - 12u^2v^6) &\leqslant M^2(3u^2-2v^2)^4 \\
\Leftrightarrow & & -3u^2\left(w^3 - (3uv^2 - 2u^3)\right)^2 + 12u^8 - 36u^6v^2 + 36u^4v^4 - 12u^2v^6 &\leqslant M^2(3u^2-2v^2)^4.
\end{aligned}$$

It suffices to prove

$$12u^8 - 36u^6v^2 + 36u^4v^4 - 12u^2v^6 \leqslant M^2(3u^2-2v^2)^4. \tag{1}$$

Note that it is natural to drop the term $\left(w^3 - (3uv^2 - 2u^3)\right)^2$ as it can be zero by the uvw theorem (and so the equality is still preserved).

If $v^2 = 0$, then we need $12u^8 \leqslant 81M^2u^8$, i.e. $M \geqslant \dfrac{2\sqrt{3}}{9}$. Otherwise, let $t = \dfrac{u^2}{v^2}$ and divide both sides of (1) by v^8. We need $12t^4 - 36t^3 + 36t^2 - 12t \leqslant M^2(3t-2)^4$, i.e.

$$\frac{12t(t-1)^3}{(3t-2)^4} \leqslant M^2.$$

(If $3t-2 = 0$, then $a^2+b^2+c^2 = 0$. This means $a = b = c = 0$, in which case the inequality is always true.) Now, let $f(t) = \dfrac{12t(t-1)^3}{(3t-2)^4}$. Then

$$f'(t) = \frac{12(t-1)^2(t+2)}{(3t-2)^5}.$$

- If $v^2 > 0$, we have $t \geqslant 1$ as $u^2 \geqslant v^2$. Then $f'(t) \geqslant 0$, so that $f(t)$ is increasing. From this, we have
$$f(t) \leqslant \lim_{t\to\infty} f(t) = \frac{4}{27},$$
which again implies $M \geqslant \dfrac{2\sqrt{3}}{9}$.

- If $v^2 < 0$, we have $t < 0$. Note that $f(t)$ is increasing on $t \in (-\infty, -2)$ and is decreasing on $t \in (-2, 0)$. So f has a maximum at $t = -2$. In other words,
$$f(t) \leqslant f(-2) = \frac{81}{512},$$
so that $M \geqslant \dfrac{9\sqrt{2}}{32}$.

By comparing the two ranges of M, we find that $M \geqslant \dfrac{9\sqrt{2}}{32}$ satisfies both inequalities. So the minimum M is $\dfrac{9\sqrt{2}}{32}$. With some computation, we see that equality holds iff $(a, b, c) = (2k, (2+3\sqrt{2})k, (2-3\sqrt{2})k)$ for some real number k or its permutation. $\square$

Problem 5.4. (Spain MO 2011 Problem 2) Let a, b, c be positive real numbers. Prove that
$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} + \sqrt{\frac{ab+bc+ca}{a^2+b^2+c^2}} \geqslant \frac{5}{2}$$
and determine when equality holds.

Solution. Let
$$\begin{cases} a+b+c &= 3u, \\ ab+bc+ca &= 3v^2, \\ abc &= w^3 \end{cases}$$
where $u, v^2, w^3 \in \mathbb{R}$. We find that
$$\begin{aligned} \sum_{\text{cyc}} a^2 &= (a+b+c)^2 - 2(ab+bc+ca) = 9u^2 - 6v^2, \\ \sum_{\text{sym}} a^2b &= (a+b+c)(ab+bc+ca) - 3abc = 9uv^2 - 3w^3, \\ \sum_{\text{cyc}} a^3 &= (a+b+c)^3 - 3\sum_{\text{sym}} a^2b - 6abc = 27u^3 - 27uv^2 + 3w^3. \end{aligned}$$
Let $\sum_{\text{cyc}} ab = k^2 \sum_{\text{cyc}} a^2$ for some $k > 0$ (so that it is easier to handle the radical). This is the same as $3v^2 = k^2(9u^2 - 6v^2)$, i.e.
$$v^2 = \frac{3k^2}{2k^2+1}u^2.$$

Note that $k \leqslant 1$ from $\sum\limits_{\text{cyc}} ab \leqslant \sum\limits_{\text{cyc}} a^2$. Now,

$$\begin{aligned}
& \frac{a}{b+c}+\frac{b}{c+a}+\frac{c}{a+b}+\sqrt{\frac{ab+bc+ca}{a^2+b^2+c^2}} \geqslant \frac{5}{2} \\
\Leftrightarrow \quad & \frac{\sum\limits_{\text{cyc}} a^3+\sum\limits_{\text{sym}} a^2b+3abc}{\sum\limits_{\text{sym}} a^2b+2abc} \geqslant \frac{5}{2}-k \\
\Leftrightarrow \quad & \frac{27u^3-18uv^2+3w^3}{9uv^2-w^3} \geqslant \frac{5}{2}-k \\
\Leftrightarrow \quad & 27u^3+\left(9k-\frac{81}{2}\right)uv^2+\left(\frac{11}{2}-k\right)w^3 \geqslant 0. \qquad (1)
\end{aligned}$$

From $k \leqslant 1$, we clearly have $\frac{11}{2}-k>0$. So (1) is linear in w^3 with positive leading coefficient. Thus, we can replace w^3 by its lower bound $3uv^2-2u^3-2\sqrt{(u^2-v^2)^3}$ by the uvw theorem. So it suffices to prove

$$27u^3+\left(9k-\frac{81}{2}\right)uv^2+\left(\frac{11}{2}-k\right)(3uv^2-2u^3-2\sqrt{(u^2-v^2)^3}) \geqslant 0,$$

i.e.

$$(2k+16)u^3+(6k-24)uv^2 \geqslant (11-2k)\sqrt{(u^2-v^2)^3}.$$

Using $v^2=\frac{3k^2}{2k^2+1}u^2$, this is the same as

$$(2k+16)u^3+\frac{3k^2(6k-24)}{2k^2+1}u^3 \geqslant (11-2k)\sqrt{\left(1-\frac{3k^2}{2k^2+1}\right)^3 u^3}.$$

Note that

$$\begin{aligned}
& (2k+16)u^3+\frac{3k^2(6k-24)}{2k^2+1}u^3 \geqslant (11-2k)\sqrt{\left(1-\frac{3k^2}{2k^2+1}\right)^3 u^3} \\
\Leftrightarrow \quad & 4k^3+32k^2+2k+16+18k^3-72k^2 \geqslant (11-2k)\sqrt{\frac{(1-k^2)^3}{2k^2+1}} \\
\Leftrightarrow \quad & (1-k)(-22k^2+18k+16) \geqslant (11-2k)\sqrt{\frac{(1-k^2)^3}{2k^2+1}}. \qquad (2)
\end{aligned}$$

From $k \leqslant 1$, we have $-22k^2+18k+16 \geqslant -22k^2+18k^2+16k^2 \geqslant 0$. So (2) is equivalent to

$$(1-k)^2(-22k^2+18k+16)^2(2k^2+1) \geqslant (11-2k)^2(1-k)^3(1+k)^3.$$

Taking out the common factor $(1-k)^2$ and expanding everything, this becomes

$$27(1-k)^2(36k^6-60k^5-9k^4+22k^3+8k^2+14k+5) \geqslant 0.$$

Finally, we have

$$
\begin{aligned}
& 36k^6 - 60k^5 - 9k^4 + 22k^3 + 8k^2 + 14k + 5 \\
&= k^4(6k-5)^2 - 34k^4 + 22k^3 + 8k^2 + 14k + 5 \\
&\geqslant -34k^4 + 22k^4 + 8k^4 + 14k^4 + 5 \\
&> 0
\end{aligned}
$$

from $k \leqslant 1$. So the inequality is true, with equality case $a = b = c$. □

The process of applying the uvw method is almost the same in all kinds of problems, i.e. changing all variables to u, v, w, using the bounds for w^3 to convert it into u, v, and proving the remaining inequality in two variables (probably one variable if it is homogeneous or is subjected to a normalized condition). We shall see one more standard example below.

Problem 5.5. Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 = 1$. Prove that

$$\frac{a^2+bc}{b+c} + \frac{b^2+ca}{c+a} + \frac{c^2+ab}{a+b} \geqslant \sqrt{3}.$$

Solution. Let

$$
\begin{cases}
a+b+c = 3u, \\
ab+bc+ca = 3v^2, \\
abc = w^3
\end{cases}
$$

where $u, v^2, w^3 \in \mathbb{R}$. We find that

$$
\begin{aligned}
\sum_{\text{cyc}} a^2 &= (a+b+c)^2 - 2(ab+bc+ca) = 9u^2 - 6v^2 = 1, \\
\sum_{\text{sym}} a^2b &= (a+b+c)(ab+bc+ca) - 3abc = 9uv^2 - 3w^3, \\
\sum_{\text{cyc}} a^2bc &= abc(a+b+c) = 3uw^3, \\
\sum_{\text{cyc}} a^2b^2 &= (ab+bc+ca)^2 - 2\sum_{\text{cyc}} a^2bc = 9v^4 - 6uw^3, \\
\sum_{\text{sym}} a^3b &= (a^2+b^2+c^2)(ab+bc+ca) - \sum_{\text{cyc}} a^2bc = 27u^2v^2 - 18v^4 - 3uw^3, \\
\sum_{\text{cyc}} a^4 &= (a^2+b^2+c^2)^2 - 2\sum_{\text{cyc}} a^2b^2 = 81u^4 - 108u^2v^2 + 18v^4 + 12uw^3.
\end{aligned}
$$

Using these identities, we see that

$$
\begin{aligned}
& \frac{a^2+bc}{b+c} + \frac{b^2+ca}{c+a} + \frac{c^2+ab}{a+b} \geqslant \sqrt{3} \\
\Leftrightarrow\quad & \sum_{\text{cyc}} a^4 + \sum_{\text{sym}} a^3b + \sum_{\text{cyc}} a^2b^2 + 4\sum_{\text{cyc}} a^2bc \geqslant \sqrt{3}\Big(\sum_{\text{sym}} a^2b + 2abc\Big) \\
\Leftrightarrow\quad & 81u^4 - 81u^2v^2 + 9v^4 - 9\sqrt{3}uv^2 + (15u + \sqrt{3})w^3 \geqslant 0.
\end{aligned}
$$

As the expression is linear in w^3 with positive leading coefficient, we apply the uvw theorem to get $w^3 \geqslant 3uv^2 - 2u^3 - 2\sqrt{(u^2-v^2)^3}$. Therefore, it suffices to prove

$$81u^4 - 81u^2v^2 + 9v^4 - 9\sqrt{3}uv^2 + (15u+\sqrt{3})(3uv^2 - 2u^3 - 2\sqrt{(u^2-v^2)^3}) \geqslant 0,$$

i.e.

$$51u^4 - 2\sqrt{3}u^3 + (-36u^2 - 6\sqrt{3}u)v^2 + 9v^4 - 2(15u+\sqrt{3})\sqrt{(u^2-v^2)^3} \geqslant 0. \qquad (1)$$

From $v^2 = \dfrac{9u^2-1}{6}$, we obtain a bound for u. Indeed, $u^2 \geqslant v^2 = \dfrac{9u^2-1}{6} > 0$ implies $\dfrac{1}{\sqrt{3}} \geqslant u > \dfrac{1}{3}$. Then (1) is the same as

$$\begin{aligned} 51u^4 - 2\sqrt{3}u^3 + (-36u^2 - 6\sqrt{3}u)\left(\frac{9u^2-1}{6}\right) \\ +9\left(\frac{9u^2-1}{6}\right)^2 - 2(15u+\sqrt{3})\sqrt{\left(u^2 - \frac{9u^2-1}{6}\right)^3} \geqslant 0. \end{aligned}$$

After some computation, we see that this means

$$69u^4 - 44\sqrt{3}u^3 + 6u^2 + 4\sqrt{3}u + 1 \geqslant \frac{2\sqrt{2}}{3}(5\sqrt{3}u+1)\sqrt{(1-3u^2)^3}.$$

Noting that $a = b = c = \dfrac{1}{\sqrt{3}}$, i.e. $u = \dfrac{1}{\sqrt{3}}$, is an equality case, we easily factorize the inequality to get

$$(\sqrt{3}u-1)(23\sqrt{3}u^3 - 21u^2 - 5\sqrt{3}u - 1) \geqslant \frac{2\sqrt{2}}{3}(5\sqrt{3}u+1)\sqrt{(1-\sqrt{3}u)^3(1+\sqrt{3}u)^3}. \quad (2)$$

Before squaring both sides, we need to ensure that $23\sqrt{3}u^3 - 21u^2 - 5\sqrt{3}u - 1 \leqslant 0$. This can be done using $u \leqslant \dfrac{1}{\sqrt{3}}$. Indeed,

$$\begin{aligned} 23\sqrt{3}u^3 - 21u^2 - 5\sqrt{3}u - 1 &\leqslant 23u^2 - 21u^2 - 5\sqrt{3}u - 1 \\ &= 2u^2 - 5\sqrt{3}u - 1 \\ &\leqslant \frac{2}{\sqrt{3}}u - 5\sqrt{3}u - 1 \\ &= -\frac{13}{3}\sqrt{3}u - 1 \\ &< 0. \end{aligned}$$

Hence, squaring both sides of (2), it is equivalent to

$$(\sqrt{3}u-1)^2(23\sqrt{3}u^3 - 21u^2 - 5\sqrt{3}u - 1)^2 \geqslant \frac{8}{9}(5\sqrt{3}u+1)^2(1-\sqrt{3}u)^3(1+\sqrt{3}u)^3.$$

After expansion, this becomes

$$(\sqrt{3}u-1)^2(19683u^6-4374\sqrt{3}u^5-729u^4+324\sqrt{3}u^3-27u^2-6\sqrt{3}u+1)\geqslant 0,$$

i.e.

$$(\sqrt{3}u-1)^2(3\sqrt{3}u-1)^4(3\sqrt{3}u+1)^2\geqslant 0.$$

(It is not hard to get this step of completing square by noting $19683=3^9$ and $4374\sqrt{3}=2\cdot 3^7\sqrt{3}$.) This inequality is clearly true. As $\frac{1}{\sqrt{3}}\geqslant u>\frac{1}{3}$, equality holds iff $a=b=c=\frac{1}{\sqrt{3}}$. □

Links

Theorems

- Muirhead's theorem
- Schur's inequality
- uvw theorem

Propositions

- 4.1: Degree 4 cyclic polynomial inequality
- 4.2: Degree 4 symmetric polynomial inequality

Problems

- 1.1: Completing squares 1
- 1.2: Completing squares 2
- 1.3: Proof of the Cauchy-Schwarz inequality
- 1.4: Proof of the AM-GM inequality (3 variables)
- 1.5: Proof of the AM-GM inequality (5 variables)
- 1.6: Expansion and completing squares
- 1.7: IMO 1999 Problem 2
- 1.8: APMO 2022 Problem 5
- 2.1: IMO HK TST 2004 Test 1 Problem 2
- 2.2: Muirhead's theorem
- 2.3: Schur's inequality
- 2.4: CWMO 2005 Problem 7
- 2.5: Tournament of Towns 1997 Spring Senior A Level Problem 5
- 2.6: Schur's inequality and homogenization
- 2.7: IMO 2005 Problem 3
- 2.8: CHKMO 2005 Problem 3
- 3.1: CGMO 2009 Problem 5
- 3.2: Proof of Schur's inequality
- 3.3: Substitution 1
- 3.4: Substitution 2
- 3.5: IMO Shortlist 2008 A7

- 3.6: IMO Shortlist 2010 A8
- 4.1: Application of degree 4 polynomial inequalities
- 5.1: IMO 2008 Problem 2
- 5.2: uvw substitution
- 5.3: IMO 2006 Problem 3
- 5.4: Spain MO 2011 Problem 2
- 5.5: uvw method

Terminologies and Notations

- uvw method

References

- Olympiad Inequalities (by Thomas J. Mildorf)
- Olympiad Inequalities — Ideas and Tricks (by Pham Văn Thuân, Lê Vĩ)
- The *uvw* method (by Mathias Bæk Tejs Knudsen)
  (Available from: https://artofproblemsolving.com/community/c6h278791p1507763)

「那都是很好很好的，可是我偏不喜歡。」

《白馬嘯西風》